John C. Borton

Data Engineer | Machine Learning Engineer

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SUMMARY

Data and software engineer specializing in high-volume data pipelines and real-time analytics, with a dual CS/Statistics background and graduate training in data analytics. Experienced designing ingestion and transformation systems for heterogeneous, high-throughput data, building automated validation frameworks, and applying statistical and machine learning methods to production and research problems.

TECHNICAL SKILLS

Data Engineering: ETL/ELT pipeline design, data wrangling & cleaning, schema/database design, relational modeling (MySQL), high-volume & heterogeneous data ingestion, real-time stream analysis

ML & Statistics: High-dimensional regression & inference, statistical simulation, feature engineering, model validation, data visualization & visual analytics

Languages: Python, R, SQL, C++, Java, JavaScript (React/Node)

Platforms & Tooling: Git, CI/CD (Jenkins), test automation frameworks, Linux, DevOps, Agile/Scrum, Jira

EXPERIENCE

Senior System / Software Test Automation Engineer, Data & Analytics — BAE Systems *2024 – Present*

- Architect data pipelines that parse, wrangle, and store high-volume, heterogeneous message formats into regularized relational databases, enabling downstream analysis across teams
- Build real-time analytics on lab hardware and flight data to score system accuracy and surface anomalies as they occur
- Develop automated testing and data-validation frameworks that verify system behavior across hardware and software subsystems, improving reliability and reducing manual QA effort
- Mentor engineers in the design and implementation of test automation and data-quality practices
- Travel to support field data analysis and present findings directly to customers

Associate Software Engineer — ASRC Federal Mission Solutions *2023 – 2024*

- Built and maintained CI/CD pipelines with Jenkins in a DevOps-oriented team, streamlining build, test, and deployment cycles
- Maintained rigorous Git version control and Linux-based workflows across remote and on-site development

Software Engineer Intern — ASRC Federal Mission Solutions *2022 – 2023*

- Contributed to production software and tooling within an Agile engineering team

RESEARCH

Statistical Simulation & Analysis (Research Assistant) — Rutgers University *2021*

- Programmed statistical simulations for Prof. Zijian Guo's research, “Doubly Debiased Lasso: High-Dimensional Inference under Hidden Confounding”
- Re-implemented the Python codebase in R, correcting errors in the original implementation and improving runtime efficiency by 150%
- Applied high-dimensional regression, confounding, and debiasing techniques to validate and extend the research findings

EDUCATION

M.S. Data Analytics — Georgia Institute of Technology*Expected May 2028*

- GPA: 4.0

B.S. Computer Science & Statistics (Dual Major), Minor: Physics*May 2023*

Rutgers University — New Brunswick, NJ

- GPA: 3.94, Summa Cum Laude · Honors Program · Phi Beta Kappa
- Graduate-level Data Interaction & Visual Analytics (completed as an undergraduate)

AWARDS

- Alan H. Grossman Scholarship — one of twenty recipients since 2011; awarded for pioneering research in Computer Science (2022)
- Eagle Scout with 3 Palms